Pemrograman 1

Full-Stack Developer
Web Development



Roni Andarsyah, ST., M.Kom

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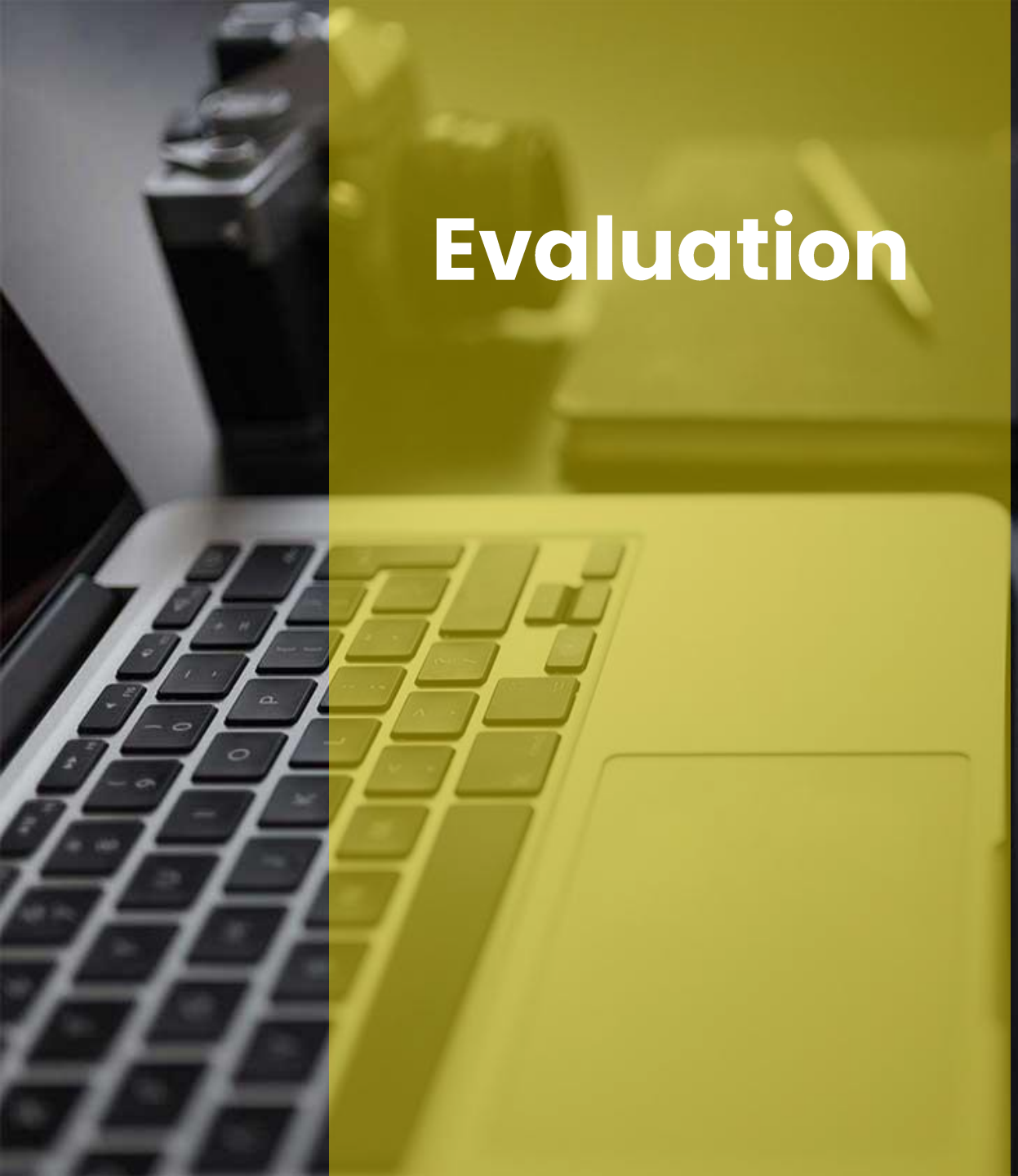
Introduction

The Complete
Web Development
Full-Stack Developer



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Evaluation

10 **Kehadiran**

4x Tidak hadir tanpa keterangan maka nilai akan otomatis E

20 **Tugas**

1x tidak mengerjakan akan mendapatkan potongan point tugas sebesar 20% otomatis

20 **ATS / UTS**

25 **PROJECT BASE FINAL**

Tugas Besar perkelompok, setiap kelompok bisa mendapatkan nilai serupa atau berbeda dengan anggota kelompoknya

25 **AAS / UAS**

SYLLABUS

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Web Development Fundamentals

1. Understand how website work and how HTML, CSS, and JavaScript contribute
2. Understand how the internet work and how websites are served
3. Learn to use git, GitHub dan Version Control
4. Learn key troubleshooting and debagging skills to aplly to your project

HTML 5

1. Learn the anatomy of HTML syntax to structure your websites.
2. Understand the HTML boilerplate and HTML doctypes. Understand indentation and nesting in HTML code.
3. Learn to use HTML tags to structure headings, paragraphs. How to structure HTML lists to create unordered and ordered lists.
4. How to insert images using HTML
5. How to create hyperlinks using anchor tags. Create multi-page websites.
6. Learn HTML best practices.

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CSS 3

1. Understand what are cascading style sheets and how you can use it to style your website.
2. How to use CSS selectors and properties.
3. Learn about how to use inline, internal and external CSS. Understand CSS coding best practices.
4. Learn about CSS specificity and inheritance.
5. Learn and understand the CSS Box Model.
6. Learn to use developer tools to inspect and diagnose CSS. Learn about CSS positioning and display properties.
7. Learn about font styling using CSS and web safe fonts. Implement responsiveness using CSS media queries. How to use CSS float and clear properties.
8. How combine CSS selectors and understand selector priority.

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FLEXBOX

1. Learn about why flexbox was introduced to CSS. Understand what happens when display is set as flex. Understand flex direction and how it affects layout. Learn to use align and justify for layout of child items
2. Understand how child items are sized and distributed in a flexbox and use the flex shorthand property.

GRID

1. Learn about how grid is different to flexbox and their strengths and weaknesses.
2. Understand what happens when display is set as grid.
3. Learn how grid sizing is done and how to create and arrange a grid.
4. Learn to position items in a grid and how placement across rows and columns is done.
5. Learn to use grid in combination with flexbox to create complex layout using native CSS. .

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BOOTSTRAP

1. Understand the difference between native CSS tooling and external frameworks.
2. Learn to install the Bootstrap framework into your website.
3. Understand the Bootstrap 12 column layout system. Learn to use Bootstrap components such as buttons, carousels, cards and navigation bars.
4. Accessing and incorporating designer icons from Bootstrap. Find and use bootstrap code snippets, examples and pre-built templates to quickly created beautiful mobile-first websites.

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WEB DESIGN

1. Learn the fundamental principles of web design, UI and UX. Understand colour theory and how to choose and use modern colour palettes to make your website look professional.
2. Understand modern typography and choose free-for-commercial use typefaces to make your website user friendly and stylish.
3. Learn to manage user attention using the key principles of
4. User Interface (UI) design.
5. Learn to design user friendly websites by understanding
6. User Experience (UX) design.

DOCUMENT OBJECT MODEL (DOM)

1. Learn the tree structure of HTML based websites. Traverse through the document using object notation. Separation of concerns and coding best practices. Manipulate and change the HTML elements using your understanding of the DOM.

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JAVASCRIPT ES 6

1. Compare programming languages like JS with scripting and markup/stylesheet languages.
2. Overview of JavaScript and its uses.
3. Understand how to work with a code editor and IDEs. Basic syntax and data types in JavaScript, including: Variables and their scope (let, const)
4. Data types (string, number, boolean, null, undefined, symbol) Operators and expressions, including arithmetic, comparison, and logical operators
5. Control structures like if/else statements and loops, including:
6. The for loop and while loop
7. The for...of loop to iterate over arrays
8. The for...in loop to iterate over object properties Functions and their importance in JavaScript, including: Function declaration and expression
9. Function scope and closures Arrow functions and their syntax Higher-order functions, including:
10. Passing functions as arguments to other functions

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JAVASCRIPT ES 6

11. Returning functions from other functions
12. The map, filter, and reduce methods on arrays and their use with higher-order functions
13. Arrays and their methods, including: Creating and accessing arrays
14. Adding and removing elements from arrays
15. The slice method for slicing arrays
16. The concat method for concatenating arrays
17. The indexOf and lastIndexOf methods for finding elements in arrays
18. The map, filter, and reduce methods for manipulating arrays The spread syntax ... to spread the elements of an array into a new array or function call
19. Object-oriented programming in JavaScript, including: Creating objects with object literals
20. Creating objects with constructor functions and the new keyword

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JAVASCRIPT ES 6

21. Creating objects with classes and the class keyword Adding and accessing properties and methods on objects The this keyword and how it works in object methods
22. The super keyword for accessing parent class methods and properties
23. Manipulating objects and arrays using methods and iteration, including:
24. The Object.keys and Object.values methods for getting the keys and values of an object as arrays
25. The for...in loop for iterating over object properties
26. The Object.assign method for copying properties from one object to another

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REACT.JS

1. Learn front-end development with React.
2. Understand when and how to use React Components.
3. Learn to pass Props and work with them.
4. Learn to write JSX and understand JSX syntax.
5. Learn about the React DOM.
6. Learn State Management in React.
7. Learn about React Hooks.
8. Learn about conditional rendering in React.
9. Understand the difference between class and functional components.

THE UNIX COMMAND LINE

1. How to use basic bash commands in a Unix/Linux Terminal.
2. How to manipulate files and folders without needing a
3. graphical user interface.
4. How to download and install to your computer using
5. command line.

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GIT GITHUB & VERSION CONTROL

1. Using git for version control and collaboration
2. Git forking, branching and cloning
3. Using GitHub as a remote repository
4. Checkout and rolling back changes with git
5. Using git and GitHub with Xcode

NODE.JS

1. Explore the components of back-end development, working with an MVC framework
2. Apply concepts like data types, objects, methods, objectoriented programming, and classes in the context of backend development.
3. Server-Side JavaScript
4. Using Node on the command line
5. NPM (Node Package Manager)
6. JavaScript Build Processes
7. Event Loop and Emitters
8. File System Interaction
9. Modules
10. Native Node drivers

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EXPRESS.JS

1. Understand how to install and use express in Node
2. applications
3. Creating Node and Express based servers
4. RESTful Routing with Express
5. Understand and use middleware for Node applications

APPLICATION PROGRAM INTERFACE (API's)

1. Understand what APIs are and how they work.
2. HTTP in Depth
3. Calling APIs
4. Reading API documentation
5. Basic API Authentication
6. Server to server communication
7. JSON vs. XML, sending data over the wire

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MYSQL / POSTGRES DATABASE

1. MySql / PostgreSQL queries
2. Create-Read-Update-Destroy (CRUD)
3. Postgres Best Practices
4. Installing Postgres
5. Relational Databases
6. Implementing One to Many, One to One and Many to Many
7. Relationships.
8. Working with Postgres in Node with the pg Package.

DEPLOYMENT

1. Understand hosting and deployment.
2. Hosting static websites with GitHub Pages.
3. Deploying server based applications with Heroku.
4. Deploying Databases with Mongo Atlas

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EJS

1. Understand what EJS does and how to use it with Node and
2. Express.
3. Templating with EJS
4. Running code in EJS templates
5. Passing data from server to template and vice versa
6. Creating layouts/partials with EJS

DATABASE FUNDAMENTAL

1. Data Relationships
2. Designing a Data Model
3. Relational Databases
4. Alternative Databases
5. Entity Relationship Modelling (ERM) and Object Relational
6. Mapping (ORM)

SQL DATABASE

1. Working with Database Schemas
2. Create-Read-Update-Destroy (CRUD)
3. Database Joins
4. Querying SQL databases

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EJS

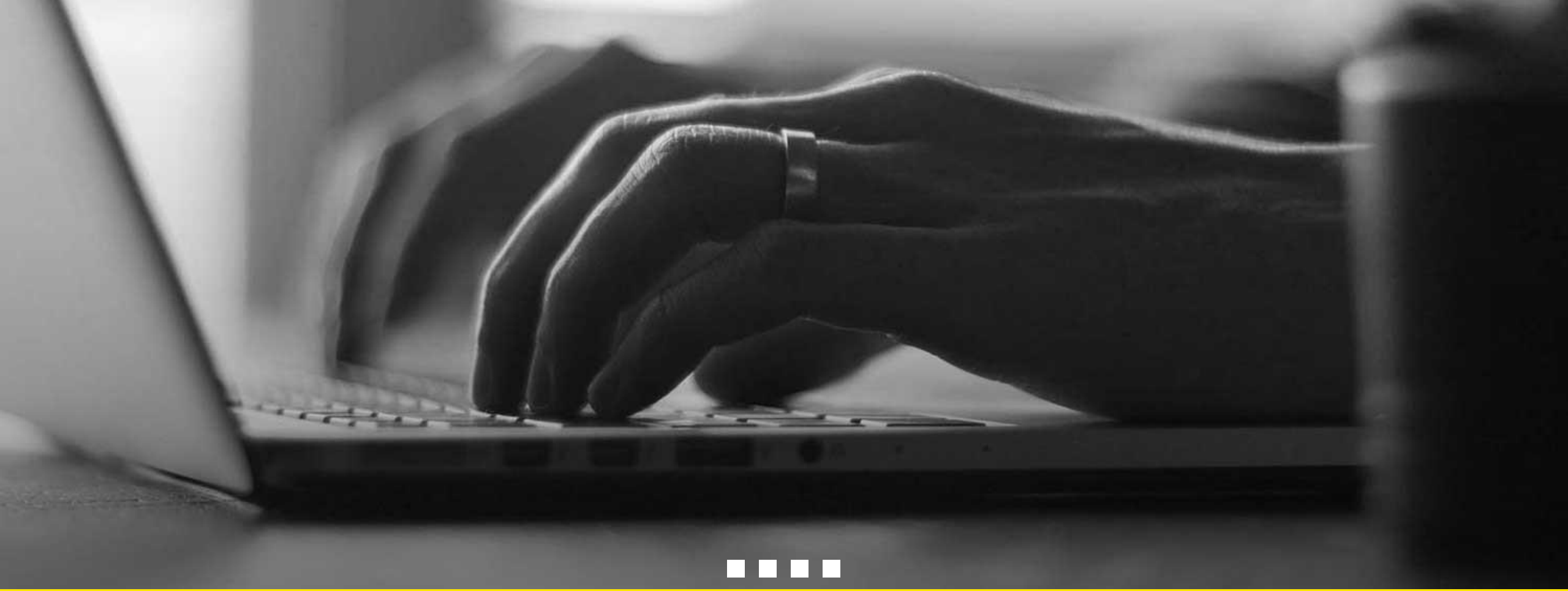
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THANK YOU

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